

Boilertronics

Sprint 1 Planning Document

Team 2: Abhi Doshi, Ethan Dixon, Ethen Chen, Keenan Koehler, Zachary Kirkeby

Sprint Overview

With this first sprint, we aim to develop the base framework upon which the game will play on. By the end of this sprint there should be a skeleton framework of functional, programmable (or not) objects, and a basic yet playable “game” with objective criteria to fulfill.

Scrum Master: Ethan Dixon

Meeting Plan:

- Tuesday, 6:30 PM
- Thursday, 6:00 PM with Project Coordinator and work time after

Risks and Challenges

As none of us have built projects using the Godot engine before, it is unlikely for us to accurately predict a story’s required work time. This, unfortunately, cannot be addressed outside of us simply gaining the experience to do so – the benefits of which will not be seen until we plan our future sprints. Consequently, it is very probable that we may have severe underestimates or overestimates.

Our inexperience with Godot may also result in inefficiencies until we become adept with the engine. As with above, this will be guaranteed until we have sufficient experience; it would be unlikely for us all to become fully acquainted with the engine within just one month/sprint, so this will likely remain a challenge for up until (at least) Sprint 3.

We are incorporating into some of our time estimates for our tasking some extra time (~30 min) to account for this learning curve and any necessary research time for the different tasking. Since we have no experience with C# and Godot, we feel this is a fair approach. Rather than dedicated R&D time, we are more focusing on research as necessary on a task by task basis as we feel this will be a more efficient use of time. Note, this time addition is strictly applied only to the first of a task of its kind, such as Story 1 T1. This time has not been applied to all stories or tasks, only a small handful based on first contact with new technologies.

Current Sprint Detail

User Story #1

As a user, I want there to be different movable, placeable elements (i.e. claws, rails, movement tiles, machines) to place on different types of gridded layers (factory layer, claw layer, and movement layer) Additionally, I want to make sure that I can only place elements where and when permitted.

#	Description	Estimated Time	Owner
1	Create class structure for placeable objects and sub-categories for placeable objects	4 hrs	Keenan
2	Create visual sprite rendering/handling system for objects	5 hrs	Ethen C.
3	Create placeholder sprites	1hrs	Ethen C
4	Create isometric layers that sense the player's mouse position	2 hrs	Keenan
5	Allow the user to drag and drop elements onto one layer	2 hrs	Keenan
6	Allow the user to interact with multiple different layers depending on what placeable element they are using	2 hrs	Keenan
7	Allow the user to move elements that they have already placed to a different spot on the same layer	1 hr	Keenan

8	Create a bitmap for each layer that will tell if the user is allowed to place elements on that space in that layer and figure out how to pass it into levels	2 hrs	Keenan
9	Create unit tests for the levels receiving bitmap information to ensure proper level construction.	2 hrs	Keenan

Acceptance Criteria:

- Given that the user is interacting with a game level, then there must exist a visible, interactable isometric grid that serves as the game's main "play area".
- Given that the user is interacting with a game level, then the user must be able to drag-and-drop objects onto the aforementioned "play area", and their placement must strictly adhere to the gridded system.
- Given that the user is interacting with a game level, then all unique visible objects must have unique visual sprites that correspond with their functionality, and also must be interactable (i.e. dragged to be re-placed elsewhere).
- Given that the user is interacting with a game level, ensure that the user can not place elements in the same layer on top of each other (excluding the claw and rails object).
- Given that the user is interacting with a game level, ensure that the user can place claws on top of rails.

User Story #2

As a developer, I want each "step" to allow the programmable objects to move.

#	Description	Estimated Time	Owner
1	Add movement commands in the scriptable elements that have the ability to move	2 hrs	Keenan
2	Allow for the elements to move via commands in their	2 hrs	Keenan

	script		
3	Reset elements that have been moved back to their original position once the user stops stepping through execution	1 hr	Keenan
4	Create unit tests that verify that the, position of the claw is in the correct position when it is commanded to move	1 hrs	Keenan

Acceptance Criteria:

- Given that the user is interacting with a game level, then programmable objects must move on synchronized “steps”.
- Given that the user is interacting with a game level, then programmable objects must be able to move in the level when given the associated command.
- Given that the user is interacting with a game level, the programmable objects must reset back to their original positions when the user presses the reset button in the UI.

User Story #3

As a developer, I want the view of the game/tiles to be isometric.

#	Description	Estimated Time	Owner
1	Make the view screen pannable	3 hrs	Ethen C.
2	Make the view screen zoomable	3 hrs	Ethen C.

Acceptance Criteria:

- Given that the user is interacting with a game level, when the user holds down the middle mouse button the view screen should be able to be panned around.
- Given that the user is interacting with a game level, then when the user uses the mouse scroll wheel, the view screen should be able to zoom in and out.
- Given that the user is interacting with a game level, then when the user uses both the middle mouse button and the scroll wheel simultaneously, then both actions should be able to occur simultaneously.

- Given that the user is interacting with a game level, then test cases for these input scenarios should be unnecessary given the relative lack of complexity and ease of testing.

User Story #8

As a user, I want to stop the level if an error or collision occurs in the user solution.

#	Description	Estimated Time	Owner
1	In the event of an element collision, a user cannot step the program until they reset the level	2 hrs	Abhi
2	In the event of a program error, the code cannot be stepped forward	2 hrs	Abhi
3	In the event of an element going out of bounds the user cannot step the program until they reset the level	2 hrs	Abhi
4	The program should identify when a movable element goes out of bounds or collides with another element	4 hrs	Abhi
5	Create unit tests for each type of error	2 hrs	Abhi

Acceptance Criteria:

- Given that the user provides programmatic instructions that lead to 2 or more of the same elements being located in the same x,y position, a collision should be noted.
- Given that the user provides programmatic instructions that lead to an element becoming out of bound, "falling off the tracks" should be noted.
- Given that an element collides in the user solution, clicking the step button should not progress the level.
- Given there is an error in the user written program, the level should be unable to step via the step button.

- Given that the step button is restricted from functioning due to a collision or code error, it should become functional once the solution is reset.

User Story #9

As a user, I want to be able to run my solution step by step.

#	Description	Estimated Time	Owner
1	Ensure the clicking of the step button progresses all code terminals by one step	2 hrs	Abhi
2	Ensure the clicking of the step button progresses the user solution (elements execute on step).	3 hrs	Abhi
3	Ensure the clicking of the step button has a visual indicator in the code editor to indicate progression through the code (visual)	1 hr	Abhi
4	Track user steps and create a counter. Reset counter for number of steps when reset button is pressed	1 hr	Abhi

Acceptance Criteria:

- Given that the user is interacting with a game level, then the user will be able to click a step button which will progress the code terminals by one singular time step.
- Given that the user is interacting with a game level, then the user will be able to click a step button which will progress the visual elements of the placeable solution by one time step.
- Given that the user is in the step mode, the line of code currently executing will be highlighted to indicate it is the current step.
- Given that the user clicks the "step" button and no error occurs, progress the step counter by one.
- Given the solution is reset, the step counter should reset to a value of zero.

- Given the step button is clicked before the step completes, the proximate clicks should not function.

User Story #10

As a user, I want to see when my solution encounters an error.

#	Description	Estimated Time	Owner
1	Developing specific error popups (vary by error)	1 hrs	Abhi
2	Enabling popups to initiate automatically from the error	2 hrs	Abhi
3	Stopping the code terminal highlighted step on the error producing line, distinguishing it from the normal display	3 hrs	Abhi
4	Displaying the error/collision within the running level UI for each type of error, no matter where in the solution it occurs	4 hrs	Abhi

Acceptance Criteria:

- Given that the user is interacting with a game level, when the user triggers an out of bounds or collision error, or other script induced error, then the user will receive a specific error popup that describes the immediate issue in the solution.
- Given that the user is interacting with a game level, then the error within the solution itself will be highlighted/depicted in a manner which allows the user to differentiate it from the rest of the UI.
- Given that the user is interacting with a game level, then in the event of a preventable, script induced error, there should exist a suggestion for avoidance of the error that will appear within a popup.
- Given that the user is interacting with a game level, then the user will be able to close the error popup message and see the error within the solution before resetting to the solution's original state before running.

User Story #11

As a developer, I want to require the user to reset their solution before being able to change their solution.

#	Description	Estimated Time	Owner
1	The solution (initial position, code, etc) must be saved before execution begins	2 hrs	Ethen C
2	The user must be unable to place any objects until the user "resets" the execution back to the initial solution.	2 hrs	Ethen C
3	Create unit tests to guarantee functionality	2 hrs	Ethen C

Acceptance Criteria:

- Given that the user is interacting with a game level, then the user's solution must be saved when the "play/step" button is pressed.
- Given that the user is interacting with a game level, then the user must be unable to place any more tiles if the solution is in "play".
- Given that the user is interacting with a game level, the user must not be able to edit any code if the solution is in "play".
- Given that the user is interacting with a game level, then when the user "resets" their solution, it must be properly restored to its state before execution.

User Story #12

As a user, I want to be able to reset my solution at any point while I'm stepping through my solution.

#	Description	Estimated Time	Owner
1	There must exist a "reset" button for the user to interact with, and expected functionality should work	3 hrs	Ethen C

2	The “reset” must not be functional while a “step” is occurring	1 hr	Ethen C
3	Create test cases simulating the user's input to ensure that the level is correctly reset	1 hr	Abhi

Acceptance Criteria:

- Given that the user is interacting with a game level, then when the user clicks the “reset” button, the “movement” layer must be reset back to before action steps begin.
- Given that the user is interacting with a game level, then when the user clicks the “reset” button, the “claw” layer must be reset back to before action steps begin.
- Given that the user is interacting with a game level, then when the user clicks the “reset” button, the “track” layer must be reset back to before action steps begin.

User Story #13

As a user, I want to be able to save and load my solutions.

#	Description	Estimated Time	Owner
1	Create a simple level saving system that saves code and tile objects; this should be saved to a temp file in the real environment.	3 hrs	Ethen C
2	Create a simple level loading system that, upon loading a save corresponding to a specific level, then the player's progress will be restored	3 hrs	Ethen C
3	Create a popup for the user to select the slot they want to save/load from	2 hrs	Ethen D.

Acceptance Criteria:

- Given that the user is interacting with a game level, then the user must be able to save a level to a local/temp file, whether this is an autosave (when exiting the level) or a manual save.
- Given that the user is interacting with a game level, then the user must be able to load a saved state for a specific level, whether this is automatically loaded (when starting a previously started level) or manually loaded.
- Given that the user is interacting with a game level, then when the user loads a save for a specific level, it must load correctly and as expected.
- Given that the user is interacting with a game level and is either saving or loading a specific level, the saves should be displayed in the format "Save Slot X". Where X is a number between 0 and 2 inclusively.
- Given that the user is interacting with a game level and is saving/loading a game, there will be a popup for the save/load option.

User Story #18

As a user, I want to have multiple sidebar terminal windows where I can write my code for the level.

#	Description	Estimated Time	Owner
1	Create a sidebar terminal window that uses a tab system	1 hrs	Zach
2	Ensure that the terminal sidebar can support multiple tabs that can be switched through and save all written code	2 hrs	Zach
3	Expanding one terminal will condense the other terminals	1 hrs	Zach

Acceptance Criteria:

- Given that the user is interacting with a game level, then there exists one sidebar terminal on the left hand side of the screen that can be expanded and condensed horizontally.
- Given that the user is interacting with a game level, there can be multiple tabs for terminals.
- Given that the user is interacting with a game level, then one can click through the tabs to get to a new tab.

- Given that the user is interacting with a game level, then when one terminal is expanded, all other opened terminals are condensed.

User Story #19

As a user, I would like to be able to write code in the in-game code editor which will be interpreted into function calls to control factory elements.

#	Description	Estimated Time	Owner
1	In the scriptable class structure, implement a mapping between interpreted commands and method calls for scriptable elements	2 hrs	Zach
2	Implement basic and dummy methods corresponding to the programming language commands	2 hrs	Zach
3	Implement a debug terminal to display level backend method calls for developer use	2 hrs	Zach
4	Interpreted commands will then be relayed as method calls to act on level elements	1 hr	Zach

Acceptance Criteria:

- There exists a debug terminal displaying the processed output from user written code for development purposes.
- Given the user is interacting with a game level, a valid script being run will result in method calls appearing in the debug terminal.
- Given the user is interacting with a game level, for each valid command in the game programming languages, there should be an equivalent method call.
- The scriptable class should contain a mapping between commands and methods.

User Story #20

As a user, I want a visual indicator that the code I have written is either correct or has some form of syntax error.

#	Description	Estimated Time	Owner
1	After code has been submitted, lines that contain unrecognized commands should be highlighted red	2 hrs	Zach
2	After the code has been submitted, malformed commands should be highlighted in red	3 hrs	Zach
3	The bottom of the terminal will list out what lines have errors	2 hrs	Zach
4	Develop unit tests that test variations of incorrect or malformed syntax, as well as cases of correct syntax	2 hr	Zach

Acceptance Criteria:

- Given that the user is interacting with a game level in the programming interface, then inputs that are not part of the programming language will be highlighted red.
- Given that the user is interacting with a game level in the programming interface, then inputs that use incorrect syntax of the programming language will be highlighted red.
- Given that the user is interacting with a game level in the programming interface, then the program will not run if there is invalid code.
- Given that the user is interacting with a game level in the programming interface, then the bottom of the terminal will list the line numbers where invalid code is detected.
- Given that the user is interacting with a game level in the programming interface, correct code will not display any errors.

User Story #21

As a developer, I want the programming language to support stoppable commands like "grab," "wait," "move."

#	Description	Estimated Time	Owner
1	Implement interpretation and support for movement based commands (move, rotate, drop, etc)	2 hrs	Zach
2	Implement interpretation and support for arithmetic commands	2 hrs	Zach
3	Implement basic registers	1 hr	Zach
4	Implement interpretation and support for basic control flow operators (jump, wait, compare, etc)	2 hrs	Zach
5	Write up initial documentation detailing currently implemented commands, command arguments, and register logic	1 hr	Zach
6.	Develop unit tests to validate the implementation of all commands and the successful operation of those commands	2 hr	Zach

Acceptance Criteria:

- Given that the user is interacting with a game level, then the “grab” command must only work for “claw” type objects, and the command must correctly command the “claw” to go through a “grab” function.
- Given that the user is interacting with a game level, then the “wait” command must work for all scriptable objects and should tell the object to wait in place for the timestep and do nothing.
- Given that the user is interacting with a game level, then the “move” command must work for “claws” and “movement” layer objects; “claws” must only move on rails in the

correct direction, and “movement” layer objects must only move/rotate rails in their own right directions.

- There exists a document detailing the current state and rules of the programming language utilized in the game levels that describes the commands, arguments, and registers that presently exist.
- Given that the user is interacting with a game level, valid arithmetic operations should be able to be performed and the results should be reflected in register states.
- Given the user is interacting with a game level, a valid script should permit use of control flow operators like compare and jump, seen by the program executing.

User Story #22

As a developer, I would like the programming language to interact with defined, scriptable objects to interact with the level in an iterative, step based process.

#	Description	Estimated Time	Owner
1	Create the scriptable interface and implement that interface on elements that we want to be scriptable (Claws, Conveyors, Rotators)	3 hrs	Keenan
2	Allow for scriptable elements to step through execution, allowing them to execute commands based on their associated script.	2 hrs	Keenan
3	Create unit tests to verify that the scriptable objects are producing the correct commands	2 hrs	Keenan

Acceptance Criteria:

- Given that the user is interacting with a game level, then a scriptable interface must exist, and must be utilized by some of the “claw” and “movement” objects.
- Given that the user is interacting with a game level, then the “claw” object must be programmable, i.e. their functionality must be scriptable.
- Given that the user is interacting with a game level, then “movement” objects must be programmable, i.e. their functionality must be scriptable.

User Story #25

As a user, I want the code editor to have different “tabs” corresponding to different elements within the level.

#	Description	Estimated Time	Owner
1	Link a given tab's interface and code to the specified object object	1 hr	Ethen C
2	Enable some highlighting system for a code editor tab's corresponding object	1 hrs	Ethen C
3	Make the scriptable objects create a new terminal when created.	1	Zach
4	Make the scriptable object delete the corresponding terminal when deleting the object.	1	Zach
5	When the script in the terminal is updated, it should change the behavior of the scriptable component according to how the script is updated. This also encompasses bug testing for the terminal to placable component relationship.	2	Zach
6	Write basic test cases and ensure functionality	2 hrs	Ethen C

Acceptance Criteria:

- Given that the user is interacting with a game level and corresponding scriptable interface exists, the “claw” objects must respond to the programmed instructions provided by the user.
- Given that the user is interacting with a game level and corresponding scriptable interface exists, the “moveable” objects must respond to the programmed instructions provided by the user.
- Given that the user is interacting with a game level and corresponding scriptable interface exists, the scriptable object should be highlighted (or otherwise indicated) when its corresponding scripting interface/tab is enabled.
- Given that the user is interacting with a game level, a terminal is created when a scriptable object is placed.
- Given that the user is interacting with a game level, the correct terminal is deleted when a scriptable object is removed.
- Given that the user is interacting with a game level, behavior of scriptable changes properly when the corresponding script is updated.

User Story #30

As a user, I want to be able to navigate through the game via menus: main to settings, main to level sections, level sections to levels.

#	Description	Estimated Time	Owner
1	Design and create a main menu screen with functional buttons.	5 hrs	Ethan D
2	Design and create a rudimentary level select screen with functional buttons.	5 hrs	Ethan D
3	Design and create a rudimentary settings screen with functional buttons.	3 hrs	Ethan D
4	Design and create a rudimentary profile screen with functional buttons.	3 hrs	Ethan D
5	Design and create a rudimentary leaderboard screen with functional buttons.	3 hrs	Ethan D

Acceptance Criteria:

- Given that the user launches the game, they should load into a functional main menu with multiple buttons.
- Given that the user is on the functional main menu, there must exist a new game button that loads the first level.
- Given that the user is on the functional main menu, there must exist a level select button that loads the level select.
- Given that the user is on the functional main menu, there must exist a settings button that loads the settings.
- Given that the user is on the functional main menu, there must exist a profile button that loads the profile screen.
- Given that the user is on the functional main menu, there must exist a leaderboard button that loads the leaderboard screen.

User Story #33

As a user, I want the level UI to be divided into different sections (Statistics, Placeable Items, Exit Menu, Time Interaction [Pause, play], etc).

#	Description	Estimated Time	Owner
1	Create placeholder buttons for “run code”, “pause”, ensure clickability	1 hrt	Ethan D
2	Create a placeholder button for “step”, “reset”, ensure clickability	1 hr	Ethan D
3	Implement an exit, level statistics, and placeable items menu.	4 hrs	Ethan D
4	Implement a UI section for code editor	1 hrs	Ethan D

Acceptance Criteria:

- Given that the user is interacting with a game level, when the user presses the level statistics button, then the level statistics menu opens.

- Given that the user is interacting with a game level, when the user presses the placeable elements button, then the placeable elements menu opens and closes.
- Given that the user is interacting with a game level, when the user presses the exit button, then the exit menu opens.
- Given that the user is interacting with a game level, when the user loads the level, there will be a currently non-functional run button.
- Given that the user is interacting with a game level, when the user loads the level, there will be a currently non-functional pause button.
- Given that the user is interacting with a game level, when the user loads the level, there will be a currently non-functional reset button.
- Given that the user is interacting with a game level, when the user loads the level, there will be a step button.

User Story #34

As a user, I want to be able to move and re-order my code editor tabs.

#	Description	Estimated Time	Owner
1	Implement functionality for the user to be able to select different tabs in the code editor.	2 hrs	Ethan D
2	Implement functionality for the user to be able to move the code editor tabs around.	2 hrs	Ethan D

Acceptance Criteria:

- Given that the user is interacting with a game level, when the user clicks a code tab, then the terminal will display the tab's code.
- Given that the user is interacting with a game level, when the user clicks and holds on the tabs, then the code editor tabs must be draggable.
- Given that the user is interacting with a game level, when dragged, then the code editor tabs must be reordered.

User Story #38

As a user, I want to be able to separate between the different layers in an isometric view.

#	Description	Estimated Time	Owner
1	Ensure that the z-index of the isometric elements ensures that proper rendering order is maintained.	2 hr	Keenan
2	Make sure that the different grid separation gives a proper feel for the layers being on separate planes, but still close enough to interact.	1 hr	Keenan
3	Ensure that elements with half height (rails, converters, rotators) still look aligned on the correct layer	1 hr	Keenan

Acceptance Criteria:

- Given the user is interacting with a game level, the elements should appear as if they are on separate planes than other elements on different layers.
- Given the user is interacting with a game level, the elements should appear as if they are on the same plane as other elements on the same layer.
- Given the user is interacting with a game level, half height elements should appear that are half the height as full sized elements.

User Story #39

As a user, I want to have the placement grid be aligned with the layer that the user is interacting with.

#	Description	Estimated Time	Owner
1	Establish a fixed height for each layer such that all layers align with the given element's corresponding layer. Ensure this is the	3 hr	Keenan

	case for all three layers the player is interacting with.		
2	Verify that the object being placed aligns with the player's mouse/where the player intends to place the object. And checking edge cases at the boundary of the coordinates for each layer.	3 hr	Keenan

Acceptance Criteria:

- Given that the user is interacting with a game level, then when the user is attempting to place a “factory” layer object, their placement grid must align with the “factory” layer.
- Given that the user is interacting with a game level, then when the user is attempting to place a “claw/rail” layer object, their placement grid must align with the “claw/rail” layer.
- Given that the user is interacting with a game level, then when the user is attempting to place a “movement” layer object, their placement grid must align with the “movement” layer.
- Given that the user is interacting with a game level, the user’s mouse position must properly relate to the coordinates on each different layer.

Total Hours Per Person for Sprint 1:

Name	Total Time
Abhi	30 hrs
Ethan D	32 hrs
Ethen C	32 hrs
Keenan	38 hrs
Zach	34 hrs

Remaining Backlog: (2025/09/26)

Functional Requirements:

- ~~1.) As a user, I want there to be different movable, placeable elements (i.e. claws, rails, movement tiles, machines) to place on different types of gridded layers (factory layer, claw layer, and movement layer). Additionally, I want to make sure that I can only place elements where and when permitted.~~
- ~~2.) As a developer, I want each "step" to allow the programmable objects to move.~~
- ~~3.) As a developer, I want the view of the game/tiles to be isometric.~~
- 4.) As a user, I would like an overarching story via a text block to better engage the user and to provide context between groups of levels.
- 5.) As a developer, I want to have a level creator framework ensuring additional levels can be created, with an interface to create levels.
- 6.) As a user, I would like a leaderboard system to compare myself to my previous scores.

- Level Functionality

- 7.) As a user, I want to be able to configure the time each 'step' takes in running my solution to the level (speed up / slow down / pause).
- ~~8.) As a user, I want to stop the level if an error or collision occurs in the user solution.~~
- ~~9.) As a user, I want to be able to run my solution step by step.~~
- ~~10.) As a user, I want to see when my solution encounters an error.~~
- ~~11.) As a developer, I want to require the user to reset their solution before being able to change their solution.~~
- ~~12.) As a user, I want to be able to reset my solution at any point while I'm stepping through my solution.~~
- ~~13.) As a user, I want to be able to save and load my solutions.~~
- 14.) As a developer, I would like a mechanism that delineates the "quality" of different interactable components in the level that have varying cost/performance factors based on their operation cost.
- 15.) As a user, I would like to be graded on my solution's statistics such as time to finish, total resources consumed, resources produced per second, and other such factors.
- 16.) As a user, I would like to be given grading "cutoffs" to have a point of reference for ideal solutions.
- 17.) As a user, I would like to be presented with unchangeable obstructions, including (but not limited to) such obstructions such as missing tiles on any layer.

- Programming Language/Editor

- ~~18.) As a user, I want to have multiple sidebar terminal windows where I can write my code for the level.~~

- 19.) ~~As a user, I would like to be able to write code in the in-game code editor which will be interpreted into function calls to control factory elements.~~
- 20.) ~~As a user, I want a visual indicator that the code I have written is either correct or has some form of syntax error.~~
- 21.) ~~As a developer, I want the programming language to support stoppable commands like "grab," "wait," "move."~~
- 22.) ~~As a developer, I would like the programming language to interact with defined, scriptable objects to interact with the level in an iterative, step-based process.~~
- 23.) As a developer, I would like the validity of the script to be checked on runtime, on a line by line basis.
- 24.) As a developer, I would like there to be dynamic error checking in the code terminal, if time allows.
- 25.) ~~As a user, I want the code editor to have different "tabs" corresponding to different elements within the level.~~
- 26.) As a developer, I would like to further extend the instruction set to support more registers and the concept of "null" if time allows.
- 27.) As a user, I would like there to be a technical manual on the in game programming language.
- 28.) As a developer, I would like there to be a gradual reveal of information pertaining to the full range and scope of the programming language to the user over the course of level progression.
- 29.) As a user, I would like to have the scripts run synchronously such that all the steps happen at the same time.

- UI/UX

- 30.) ~~As a user, I want to be able to navigate through the game via menus: main to settings, main to level sections, level sections to levels.~~
- 31.) As a user, I would like a quick reference panel / menu within levels for global hints based on progress.
- 32.) As a user, I would like a hint system that would allow the user to see the quick reference for the concept introduced in the level.
- 33.) ~~As a user, I want the level UI to be divided into different sections (Statistics, Placeable Items, Exit Menu, Time Interaction [Pause, play], etc).~~
- 34.) ~~As a user, I want to be able to move and re-order my code editor tabs.~~
- 35.) As a user, I want the floor layer to have default visuals and each of the placeable elements that can be put on the floor layer to have separate default visuals too.
- 36.) As a user, I want each placeable element on the movement layer to have visuals beyond a baseline placeholder sprite.
- 37.) As a user, I want each placeable element on the claw layer to have visuals beyond a baseline placeholder sprite.
- 38.) As a user, I want to be able to separate between the different layers in an isometric view.

- ~~39.) As a user, I want to have the placement grid be aligned with the layer that the user is interacting with.~~
- 40.) As a user, I want to toggle the view of a specific layer on and off.
 - 41.) As a user, as I am placing tiles, I want the tiles on the corresponding layer to be more opaque than others for visibility improvements.
 - 42.) As a user, I want some sprites (claws moving up/down/left/right, machines working/creating items, and movement layers operating) to be animated when that element performs some action to give the user visual feedback.
 - 43.) As a developer, I want trains to connect the levels in the level select menu.
 - 44.) As a user, I would like sound effects to accompany actions from interactable game mechanisms, time iteration, and other mechanics.
 - 45.) As a user, I want to be able to change the game's volume settings.

- **Server Integration**

- 46.) If time allows, as a user, I want to be able to create and login to an account.
- 47.) If time allows, as a user, I want to be able to play the game on Steam
- 48.) If time allows, as a user, I would like to be able to see other people's scores online and compare my score to theirs.
- 49.) If time allows, as a user, I would like to view the score my friends got online and compare my scores to theirs.
- 50.) If time allows, as a user, I would like to design my own levels and share them online.

- **Extras**

- 51.) As a user, I would like an achievements system derived from game and challenge completion.
- 52.) As a developer, I want a level editing system to make designing new levels easier.
- 53.) If time allows, as a developer, I would like to include easter eggs into the game.
- 54.) If time allows, as a user, I would like a block mode for the scripts rather than a strictly terminal based approach.
- 55.) If time allows, as a user, I would like to have a modding framework.
- 56.) If time allows, as a user, I would like more levels with different gimmicks such as pipes, electricity, heat, and other similar mechanics.
- 57.) If time allows, as a user, I would like to have colorblind modes.

Non-functional Requirements:

- 58.) As a user, I want good-looking tile sprites.
- 59.) As a user, I want responsive and intuitive buttons.
- 60.) As a user, I would like the game's visuals to be appealing and easy to read.
- 61.) As a user, I want to be able to scale the UI to make it easier to read.
- 62.) As a user, I want the animation for the given action I programmed to execute within 5 seconds of the game indicating that it should be executing. (performance)

- 63.) If time allows, as a user, I would like my data to be handled securely over the network when storing their solution metrics by utilizing HTTPS. (security)
- 64.) If time allows, as a developer, I want the user's data to be secure and not accessible by unauthorized parties.

Gantt Chart Below

[illegible]

Task ID	Task Description	Assignee	Start Date	End Date	Progress
30.1	Design and create a main menu screen with functional buttons.	Ethan D	9/29/25	10/5/25	100%
30.2	Design and create a rudimentary level select screen with functional buttons.	Ethan D	9/29/25	10/5/25	100%
30.3	Design and create a rudimentary settings screen with functional buttons.	Ethan D	9/29/25	10/5/25	100%
30.4	Design and create a rudimentary profile screen with functional buttons.	Ethan D	10/4/25	10/12/25	100%
30.5	Design and create a rudimentary leaderboard screen with functional buttons.	Ethan D	10/4/25	10/12/25	100%
33 Level UI					
33.1	Create placeholder buttons for "run code", "pause", ensure clickability	Ethan D	10/1/25	10/6/25	100%
33.2	Create a placeholder button for "step", "reset", ensure clickability	Ethan D	10/1/25	10/6/25	100%
33.3	Implement an exit, level statistics, and placeable items menu.	Ethan D	10/1/25	10/6/25	100%
33.4	Implement a UI section for code editor	Ethan D	10/1/25	10/6/25	100%
34 Move and re-order code editor tabs					
34.1	Implement functionality for the user to be able to select different tabs in the code editor.	Ethan D	10/8/25	10/17/25	100%
34.2	Implement functionality for the user to be able to move the code editor tabs around.	Ethan D	10/8/25	10/17/25	100%
38 Separatable layers in isometric view					
38.1	Ensure that the z-index of the isometric elements ensures that proper rendering order is maintained.	Keenan	10/6/25	10/12/25	100%
38.2	Make sure that the different grid separation gives a proper feel for the layers being on separate planes, but still close enough to interact.	Keenan	10/6/25	10/12/25	100%
38.3	Ensure that elements with half height (rails, converters, rotators) still look aligned on the correct layer	Keenan	10/6/25	10/12/25	100%
39 Placement grid should be aligned with the user's active grid					
39.1	Establish a fixed height for each layer such that all layers align with the given element's corresponding layer. Ensure this is the case for all three layers the player is interacting with.	Keenan	10/11/25	10/17/25	100%
39.2	Verify that the object being placed aligns with the player's mouse/where the player intends to place the object. And checking edge cases at the boundary of the coordinates for each layer.	Keenan	10/11/25	10/17/25	100%